

Independent Section

Contains tests that are independent of the class of modeled organism, a model's complexity or types of identifiers that are used to describe its components. Parameterization or initialization of the network is not required. See readme for more details.

Consistency

Stoichiometric Consistency	100.0%	
Mass Balance	96.4%	
Charge Balance	100.0%	
Metabolite Connectivity	100.0%	
Unbounded Flux In Default Medium	21.8%	

Sub Total	88%	
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Annotation - Metabolites

Presence of Metabolite Annotation	100.0%	
Metabolite Annotations Per Database	Info	
pubchem.compound	70.3%	
kegg.compound	49.1%	
seed.compound	41.4%	
inchikey	0.0%	
inchi	61.7%	
chebi	42.6%	
hmdb	56.1%	
reactome	34.9%	
metanetx.chemical	73.7%	
bigg.metabolite	100.0%	
biocyc	43.8%	
Metabolite Annotation Conformity Per Database	Info	
pubchem.compound	100.0%	
kegg.compound	95.9%	
seed.compound	80.7%	
inchikey	0.0%	
inchi	98.9%	
chebi	100.0%	
hmdb	23.0%	
reactome	35.0%	
metanetx.chemical	100.0%	
bigg.metabolite	100.0%	
biocyc	94.9%	
Uniform Metabolite Identifier Namespace	0.0%	

Sub Total	57%	
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Annotation - Reactions

Presence of Reaction Annotation	100.0%	
Reaction Annotations Per Database	Info	
rhea	14.7%	
kegg.reaction	12.6%	
seed.reaction	0.0%	
metanetx.reaction	62.7%	
bigg.reaction	99.7%	
reactome	9.0%	
ec-code	16.8%	
brenda	0.0%	
biocyc	14.0%	
Reaction Annotation Conformity Per Database	Info	
rhea	85.3%	
kegg.reaction	99.8%	
seed.reaction	0.0%	
metanetx.reaction	100.0%	
bigg.reaction	100.0%	
reactome	91.6%	
ec-code	100.0%	
brenda	0.0%	
biocyc	99.1%	
Uniform Reaction Identifier Namespace	100.0%	

Sub Total	75%	
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Annotation - Genes

Presence of Gene Annotation	100.0%	
Gene Annotations Per Database	Info	
refseq	99.2%	
uniprot	99.0%	
ecogene	0.0%	
kegg.genes	0.0%	
ncbigi	0.0%	
ncbigene	100.0%	
ncbiprotein	0.0%	
ccds	98.1%	
hprd	0.0%	
asap	0.0%	
Gene Annotation Conformity Per Database	Info	
refseq	48.3%	
uniprot	62.9%	
ecogene	0.0%	
kegg.genes	0.0%	
ncbigi	0.0%	
ncbigene	100.0%	
ncbiprotein	0.0%	
ccds	3.3%	
hprd	0.0%	
asap	0.0%	

Sub Total	54%	
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Annotation - SBO Terms

Metabolite General SBO Presence	100.0%	
Metabolite SBO:0000247 Presence	100.0%	
Reaction General SBO Presence	100.0%	
Metabolic Reaction SBO:0000176 Presence	100.0%	
Transport Reaction SBO:0000185 Presence	100.0%	
Exchange Reaction SBO:0000627 Presence	100.0%	
Demand Reaction SBO:0000628 Presence	100.0%	
Sink Reactions SBO:0000632 Presence	100.0%	
Gene General SBO Presence	100.0%	
Gene SBO:0000243 Presence	100.0%	
Biomass Reactions SBO:0000629 Presence	100.0%	

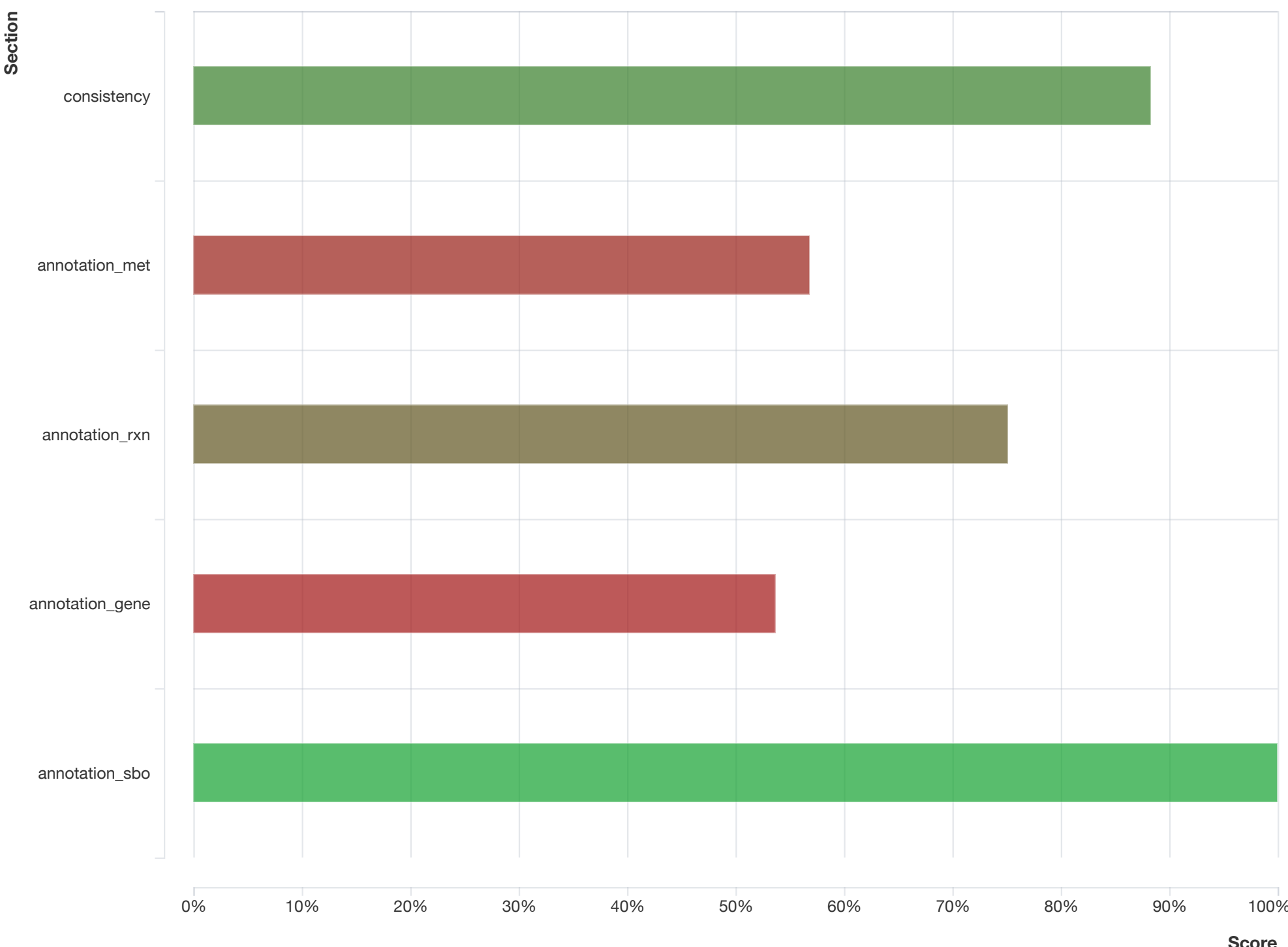
Sub Total	100%	
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Total Score	88%	
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Total Score

88%

Score per Category



Specific Section

Covers general statistics and specific aspects of a metabolic network that are not universally applicable. See readme for more details.

SBML

SBML Level and Version	SBML Level 3 Version 1	
FBC enabled	true	

Basic Information

Model Identifier	iMiceBrain	
Total Metabolites	3,111	
Total Reactions	6,042	
Total Genes	1,668	
Total Compartments	9	
Metabolic Coverage	3.62	
Unconserved Metabolites	0	
Minimal Inconsistent Net Stoichiometries	0	

Metabolite Information

Unique Metabolites	3,111	
Duplicate Metabolites in Identical Compartments	12	
Metabolites without Charge	0	
Metabolites without Formula	0	
Medium Components	293	

Reaction Information

Purely Metabolic Reactions	2,985	
Purely Metabolic Reactions with Constraints	22	
Transport Reactions	2,658	
Transport Reactions with Constraints	11	
Reactions With Partially Identical Annotations	0.20	
Duplicate Reactions	0.00	
Reactions With Identical Genes	0.63	

Gene-Protein-Reaction (GPR) Associations

Reactions without GPR	1,150	
Fraction of Transport Reactions without GPR	0.30	
Enzyme Complexes	0	

Biomass

Biomass Reactions Identified	1	
Biomass Consistency	Errored	
Biomass Production In Default Medium	176.30	
Unrealistic Growth Rate In Default Medium	true	
Biomass Production In Complete Medium	410.30	
Blocked Biomass Precursors In Default Medium	0	
Blocked Biomass Precursors In Complete Medium	0	
Ratio of Direct Metabolites in Biomass Reaction	0.00	
Number of Missing Essential Biomass Precursors	14	

Energy Metabolism

Non-Growth Associated Maintenance Reaction	Errored	
Growth-associated Maintenance in Biomass Reaction	false	
Number of Reversible Oxygen-Containing Reactions	11	
Erroneous Energy-generating Cycles	Info	
MNXM3	Errored	
MNXM63	Errored	
MNXM51	Errored	
MNXM121	Errored	
MNXM423	Errored	
MNXM6	0	
MNXM10	0	
MNXM36	Skipped	
MNXM208	Skipped	
MNXM191	Skipped	
MNXM223	Skipped	
MNXM7517	Skipped	
MNXM12233	Skipped	
MNXM558	Skipped	
MNXM21	Errored	
MNXM89557	Errored	

Network Topology

Universally Blocked Reactions	0	
Orphan Metabolites	0	
Dead-end Metabolites	0	
Stoichiometrically Balanced Cycles	2,118	
Metabolite Production In Complete Medium	536	
Metabolite Consumption In Complete Medium	536	

Matrix Conditioning

Ratio Min/Max Non-Zero Coefficients	0.00	
Independent Conservation Relations	69	
Rank	2918	
Degrees Of Freedom	3124	

Experimental Data Comparison

Growth Prediction	Skipped	
Gene Essentiality Prediction	Skipped	

Misc. Tests

Environment

Python Version	3.10.5
Platform	Darwin
Memote Version	